Build a feature matrix and workflow demo for Semester Faculty OS — executive assistant, research partner, and AI teaching assistant capabilities

Computer
Turn Semester Intelligence's teacher OS features into a deployment playbook — automated grading, lecture planning, and administrative assistant workflows
Computer
Detail the executive assistant mode for scheduling and admin
Map out how the research partner mode handles data privacy
Design a concrete workflow for the AI teaching assistant

Semester Faculty OS should combine three coordinated intelligence modes: an executive assistant for scheduling and administration, a research partner for evidence-based scholarship, and an AI teaching assistant for course support. The deployment principle is **prepare → review → authorize → execute → reconcile**: automate preparation and permitted routine work, while preserving human authority over grades, publication, sensitive data, and consequential decisions.

Below is a feature matrix, four concrete workflow demos, and a deployment playbook. These are proposed specifications, not claims of implementation or deployment. Your existing Course Studio prototype supports the direction, but the AI documents contain conflicting testing claims: the routing document says provider exclusion is red-team tested, while the platform maturity document says the red-team suite has not run. Treat that control as unverified until an actual staging test report resolves the conflict. [1] [2] [3]

1. Faculty OS feature matrix

Use one Faculty OS and one policy-controlled intelligence gateway, with separate permissions and source collections for each mode. The existing role model already requires identity, membership, capability, resource relationship, consent, classification, policy, entitlement, and workflow state to participate in access decisions. [4]

Mode / feature	Inputs permitted by design	Prepared output	Execution boundary	Required evidence
Executive: daily briefing	Authorized calendar, tasks, course deadlines, work queues	Source-linked priorities and blockers	No automatic changes to commitments	Source freshness and permission tests

Mode / feature	Inputs permitted by design	Prepared output	Execution boundary	Required evidence
Executive: schedule planning	Availability, preferences, fixed commitments	Proposed time blocks and alternatives	Exact calendar changes require approval or a narrowly approved standing rule	Conflict, retry, revocation tests
Executive: meeting preparation	Authorized agenda and documents	Briefing, questions, draft agenda	Recording and sharing require separate permission	Document-scope and recipient tests
Executive: communications	Approved course/admin context	Announcement, email, reminder drafts	Review recipients and content before sending	Approval binding and duplicate-send protection
Executive: administrative forms	Authorized evidence and approved templates	Prefilled draft with missing fields flagged	Human signs or submits consequential forms	Provenance and validation tests
Research: literature review	Public or licensed sources the researcher may access	Evidence matrix and synthesis	No claim of reading unavailable full text	Citation and extraction evaluation
Research: methods support	Authorized protocol and project context	Assumptions, alternatives, critique	Researcher owns methodological decisions	Expert review and uncertainty handling
Research: computation	Approved dataset and code in a permitted environment	Code, diagnostic reports, executed results	Execution permissions separate from code drafting	Sandbox, reproducibility, data-egress tests
Research: writing	Authorized sources and manuscript	Draft, revision plan, claim/source checks	No autonomous publication or submission	Citation and confidentiality checks
AI TA: course Q&A	Student-visible approved course sources	Cited explanation or logistics answer	Cannot retrieve instructor-only answer keys	Source-isolation and integrity tests
AI TA: practice support	Approved concepts and synthetic exercises	Hints, questions, worked practice where permitted	Assignment policy governs help	Refusal and pedagogical evaluation
AI TA: faculty preparation	Instructor-approved materials	Lecture plan, practice questions, FAQ drafts	Faculty approves publication	Accessibility and source checks
Assessment: deterministic scoring	Authorized answers, tests, scoring rules	Computed provisional score and test results	Official release remains governed	Reproducibility and scoring-rule tests
Assessment: AI feedback assistance	Only data permitted by the effective AI policy	Proposed feedback or rubric annotations	No independent official grade decision	Privacy approval, evaluation, human review

The last two rows must remain distinct. Deterministic scoring is not the same as model-generated grading, and neither automatically authorizes grade release.

2. Executive assistant: scheduling and administration

Control model

The executive assistant should offer three levels:

1. Read and prepare: briefings, plans, drafts, conflict detection.
2. Approved action: execute an exact calendar change or communication after review.
3. Bounded standing permission: perform narrowly defined routine actions under a revocable rule.

A standing rule might permit creating private research blocks in free time. It should not authorize moving classes, changing assessment deadlines, cancelling student appointments, or sending messages.

Workflow demo: prepare the faculty week

Illustrative request:

> "Protect two 90-minute research blocks, prepare Thursday's lecture, and organize grading. Do not move teaching or office hours."

Step	System behavior	Screen / artifact
1. Resolve context	Identify faculty, institution, active courses, authorized calendars	Context header
2. Read constraints	Retrieve fixed events, due dates, task status, preferences	Evidence drawer
3. Detect conflicts	Identify collisions, insufficient free time, stale external calendars	Conflict list
4. Propose plan	Offer research blocks and preparation/grading tasks	Weekly plan preview
5. Explain tradeoffs	State which work remains unscheduled and why	Assumption panel
6. Review action	Show exact calendars, times, titles, visibility, attendees	Approval screen
7. Revalidate	Check availability and permission again before writing	Execution status
8. Execute once	Apply approved changes with duplicate protection	Action receipt
9. Reconcile	Verify returned events and refresh the faculty view	Confirmed schedule

Illustrative output

```
Proposed:
- Tuesday 9:00-10:30: private research block
- Wednesday 14:00-15:30: private research block
- Wednesday 11:00-11:45: lecture preparation task
- Friday 10:00-11:00: grading review task

Preserved:
- Teaching sessions
- Office hours
- Student appointments
```

```
Not yet changed:
- No invitations sent
- No course deadlines moved
- No events written before approval
```

If the calendar changes between preview and execution, stop and refresh the proposal. Do not silently choose another time.

Administrative workflow demo: committee meeting

```
Meeting selected
  → retrieve permitted agenda and documents
  → prepare source-linked briefing
  → faculty edits agenda/questions
  → optional approved distribution
  → faculty confirms meeting decisions
  → draft follow-up tasks
  → approve owners and deadlines
  → create tasks and verify results
```

Important controls:

- A generated summary is not an official committee decision.
- Task owners are proposed, not assigned without appropriate authority.
- Recording/transcription requires an explicitly permitted workflow.
- Student, personnel, and confidential committee records do not become general assistant memory.
- Messages must preview recipients, attachments, and the complete content.

This extends the existing shared workflow and approval architecture rather than adding a second automation system.[5] [4]

3. Research partner: privacy architecture

Research privacy needs more than a “private project” toggle. Each source, dataset, collaborator, processing environment, and model invocation needs an explicit access and use decision.

Proposed data classes

Class	Examples	Default research-partner behavior
Public	Open publications, public datasets	Retrieve and analyze with provenance
Licensed	Subscription articles and licensed datasets	Use only within verified access and license constraints
Confidential	Unpublished manuscripts, collaborator documents	Restrict project access; use only approved processing environments

Class	Examples	Default research-partner behavior
Restricted participant data	Identifiable interviews, participant records, sensitive datasets	External-model processing blocked by default
Student-linked institutional records	Submissions, educational records, advising information	Do not send to AI providers under the current project rule
Synthetic / approved de-identified	Synthetic examples or reviewed transformations	Use only after classification and intended-use approval

The existing routing and maturity specifications prohibit sending student records to providers. That boundary remains in force; faculty consent or an assistant setting does not override it.[2] [3]

Privacy workflow

```
Source or dataset added
  → establish ownership and permitted purpose
  → classify data
  → identify institutional/protocol/license restrictions
  → approve project and collaborator scope
  → select permitted storage and compute environment
  → authorize retrieval and tool use
  → minimize request context
  → inspect outbound payload
  → process only if policy allows
  → label output and provenance
  → apply retention, access review, and closure rules
```

Privacy controls to implement

Control	Required behavior
Project isolation	Membership in one project does not grant access to another
Purpose limitation	Approved teaching use does not imply research reuse
Source permissions	Retrieval checks access before returning passages
Index isolation	Search indexes and embeddings inherit source restrictions
Egress control	Check tool arguments, files, retrieved passages, and prompts—not only user text
Provider control	Apply approved provider, retention, region, and contractual requirements
Memory control	No cross-project learning or persistent sensitive memory by default
Output classification	Derived artifacts inherit appropriate source restrictions
Logs	Record operational evidence without retaining unnecessary sensitive payloads
Revocation	Remove access from files, search, caches, tools, and future runs
Retention	Respect approved schedules and holds; report deletion limitations accurately

Removing names alone is not sufficient evidence that a dataset is safe to export. Require an approved transformation and re-identification review appropriate to the dataset and use.

Research demo: interview study

Illustrative request:

> "Analyze recurring themes in these participant interviews."

The safe workflow is:

1. Detect that the project contains participant information.
2. Block external model transmission by default.
3. Display the applicable project and institution restrictions.
4. Offer permitted alternatives:
    - Manual coding assistance without accessing the interviews.
    - Synthetic examples for designing the coding framework.
    - Analysis in an explicitly approved controlled environment, if one exists.
5. Have the researcher approve the coding framework.
6. Produce traceable annotations where permitted.
7. Keep quotations linked to restricted source records.
8. Require researcher review before dissemination.

An internal or locally hosted model is not automatically authorized simply because it is not an external provider. It still requires access, infrastructure, protocol, retention, and security approval.

Faculty-facing privacy receipt

Each research run should show:

```
Project and purpose
Data classification
Sources accessed
Processing environment
Provider/model, if used
Context transmitted
Retention behavior
Output classification
Execution status
Reviewer / approval reference
```

The routing document already calls for showing which provider/model was selected, what context was shared, sources used, retention, cost, and fallback behavior.[2]

4. AI TA: concrete teaching workflow

Demonstration course

Use a synthetic introductory operating-systems course for the demo.

Approved student-visible sources:

- Published syllabus.
- Instructor-approved process-management notes.
- A practice fork/exec/wait example.
- Published assignment instructions.

Excluded student-facing sources:

- Graded-assignment solution.
- Instructor-only test cases.
- Answer keys.
- Other students' submissions.
- Private student notes and conversations.

The Course Studio prototype already supports assignment-level AI modes such as Off, Explain only, and Feedback on drafts. Its policy hierarchy prevents course rules from being more permissive than institution rules.[1]

Student request

> "I don't understand why my child process becomes a zombie. Can you explain it?"

Workflow

1. Authenticate the student and resolve the assigned course.
2. Determine whether the request concerns practice or assessed work.
3. Load effective institution, course, and assignment rules.
4. Retrieve only student-visible approved sources.
5. Explain the concept with citations to those course sources.
6. Ask a diagnostic question:
   > "Where does the parent collect the child's exit status?"
7. Offer a hint or approved practice exercise.
8. If the student requests a prohibited complete solution, refuse that portion and offer permitted conceptual help.
9. If the sources conflict or the policy is unclear, escalate rather than invent an answer.
10. Store only the permitted interaction metadata and student-controlled history.

Faculty review workflow

```
Unresolved question
  → minimum-necessary escalation
  → course question queue
  → instructor review
  → drafted clarification
  → source / policy / accessibility review
  → approved publication
  → updated FAQ and student notification
```

Faculty should receive a permitted question or aggregate theme, not automatic access to private student conversations. Any shared transcript or excerpt requires an explicit, policy-compliant sharing basis.

Acceptance tests

- A student request cannot retrieve instructor-only answer keys.
- Instructions embedded in retrieved documents cannot expand tool permissions.
- Turning AI off blocks invocation, not just the UI.
- Explain-only mode does not produce a complete assessed solution.
- Unsupported claims trigger uncertainty or escalation.
- Expired course membership blocks retrieval.
- A provider outage produces a clear fallback, not fabricated completion.
- Faculty summaries do not expose private conversations.
- Published clarifications preserve source and version history.

5. Automated grading: three separate capabilities

“Automated grading” should not be a single feature flag.

Capability	What can be automated	Release boundary
Deterministic scoring	Answer-key comparisons, instructor-defined calculations, bounded code tests	Provisional scores until authorized release
AI-assisted review	Feedback drafts, rubric suggestions, consistency review	Human accepts, edits, or rejects; privacy policy must permit processing
Official grade release	Approved batch execution and passback	Authorized faculty decision, audit, receipt, reconciliation

The current lifecycle specifies assessment, moderation where required, release, and regrade; controlled-action patterns give grade release to the faculty of record and preserve the external authoritative system during transition.[4] [5]

Deterministic code-lab demo

```
Submission received
  → receipt and immutable submission version
  → validate permitted file types
  → execute in an isolated, resource-limited sandbox
  → run versioned instructor-approved tests
  → separate student failure from infrastructure failure
  → calculate provisional rubric score
  → human review of flagged or disputed cases
  → approve exact release batch
  → publish / pass back
  → reconcile
  → offer regrade path
```

Required safeguards:

- No production credentials or institutional records in the runner.
- Network blocked unless the assignment explicitly requires approved access.
- CPU, memory, process, storage, and runtime limits.
- Versioned test suite and scoring formula.
- Timeout is not automatically interpreted as academic misconduct.
- Infrastructure failures do not become student penalties.
- Partial grade passback remains visible and owned.
- Reruns preserve the original submission and prior results.

AI feedback restriction

Do not enable provider-based review of identifiable student submissions under the current “no student records to providers” rule. Launch AI feedback demonstrations on synthetic work first. Any broader processing design requires a separately approved policy, architecture, and evidence package. [3] [2]

6. Lecture-planning demo

Faculty request:

> “Prepare a 50-minute lesson on process lifecycle using my approved notes. Include a misconception check and a short practice activity.”

Proposed output:

Segment	Duration	Purpose
Opening scenario	5 minutes	Establish the problem
Process lifecycle explanation	12 minutes	Connect creation, execution, termination, and reaping
Worked example	13 minutes	Trace parent and child behavior

Segment	Duration	Purpose
Practice activity	12 minutes	Apply the model
Misconception check	5 minutes	Identify unresolved understanding
Wrap-up	3 minutes	Summarize and point to practice

Semester should also prepare:

- Objective-to-activity mapping.
- Source references.
- Instructor notes.
- Student-visible materials.
- Alternative explanations.
- Accessibility checks.
- A student preview.
- A publication-impact preview.

Faculty reviews and approves before publication. Instructor-only solutions must stay outside the student-visible source collection.

7. Deployment playbook

These stages are proposed activation gates, not a claim that the Faculty OS is ready to deploy.

Stage	Work	Owner	Exit evidence
0. Reconcile	Map existing screens, commands, policies, source collections, tests	Product + engineering	No duplicate workflow or conflicting source of truth
1. Security foundation	Verify tenant/course/project scope, invocation policy, revocation, egress controls	Platform + security	Positive and negative authorization tests
2. Synthetic demonstrations	Run scheduling, lecture, literature, AI TA, and scoring scenarios	Product + faculty reviewers	Reproducible demo results
3. Shadow evaluation	Prepare outputs without publishing, sending, or changing official records	AI governance + domain owners	Quality, privacy, integrity, and failure review
4. Assist mode	Allow drafts and source-grounded support for approved cohorts	Institution + faculty owners	Training, support, monitoring, accepted evaluations
5. Bounded actions	Enable selected calendar writes and publication commands	Domain owner + release owner	Exact approval, idempotency, receipt, reconciliation
6. Assessment operations	Enable authorized deterministic scoring and grade-release workflow	Faculty + registrar/academic owner	Scoring acceptance, sandbox tests, correction/regrade path
7. Expansion	Add approved projects, courses, and integrations	Institution + implementation	Evidence reviewed per scope; rollback remains available

Resolve the evidence conflict first

Before activation, reconcile:

- The AI routing document's claim that student-record provider exclusion is red-team tested.
- The maturity document's statement that the red-team suite has not run.
- The earlier architecture snapshot's warning that tenant AI policy was not enforced at the call.

Require a dated test report tied to the deployed gateway version. A policy statement, prototype notice, or library test is not enough. [6] [3] [2]

Kill switches and recovery

Provide separate controls for:

- All model invocations.
- External-provider egress.
- Calendar and messaging writes.
- Course publication.
- Research computation.
- Assessment runners.
- Grade-release commands.

Stopping AI should not disable access to course materials, existing records, human support, or already-released feedback.

8. Launch acceptance checklist

- [ ] All three modes use the same gateway and authorization model.
- [ ] Drafting permission is separate from execution permission.
- [ ] Institution, course, assignment, and project policies are evaluated at invocation.
- [ ] Private student work is excluded from faculty retrieval by default.
- [ ] Instructor-only assessments are isolated from student-facing retrieval.
- [ ] Research permissions cover storage, indexes, tools, logs, outputs, and retention.
- [ ] Scheduling actions handle stale availability and retries safely.
- [ ] Communications bind exact recipients, attachments, and content.
- [ ] Deterministic scoring is versioned and reproducible.
- [ ] AI-generated feedback never independently becomes an official grade.
- [ ] Grade writes produce confirmed receipts and reconciliation status.
- [ ] Critical workflows pass accessibility and failure-path tests.

- [ ] Each capability has a domain owner, support runbook, monitoring, and activation approval.
- [ ] Testing claims correspond to actual reports for the released version.

The most useful first acceptance demo is a connected teaching chain: the executive assistant schedules preparation, the research partner verifies teaching sources, the AI TA drafts a policy-compliant explanation, the faculty member publishes it, and an authorized assessment workflow later produces reviewed, reconciled feedback. That proves the modes operate as one system without collapsing their privacy or authority boundaries.

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1. The Main Semester design system (2) copy/handoff/design_handoff_permissions_and_queues/design/templates/CourseStudio.dc.html
2. The Main Semester design system (2) copy/docs/system/algorithms/SEMESTER_AI_ROUTING.md
3. The Main Semester design system (2) copy/docs/platform/AI_INTELLIGENCE_PLATFORM.md
4. The Main Semester design system (2) copy/docs/system/ROLE_LIFECYCLES.md
5. The Main Semester design system (2) copy/docs/operations/CONTROLLED_ACTION_PATTERNS.md
6. The Main Semester design system (2) copy/uploads/Semester build Architecture.pdf